

```

root_path = fileparts(mfilename("fullpath"));
root_path = pwd;
data_path = fullfile(root_path, 'sym4D_cutsAndFits');

if ~exist('cuts2fit400', 'var')
    ld = load(fullfile(data_path, 'multicuts_fit_dataGH_ei400meV.mat'));
    cuts2fit400 = ld.cuts2fit400;
end
if exist('used_ds400', 'var')
    used_ds = used_ds400;
else
    used_ds = containers.Map()
    sel_info = struct('cuts', [], 'en_range', [], 'fname', '', 'bg_par', []);
    %
    sel_info.cuts = cuts2fit400.cuts_dir001FFfit(1);
    sel_info.en_range = 50:10:160;
    sel_info.cut_dim = 2;
    sel_info.fname = 'EnFit_Ei400ref001off110_2D_constFFCut';
    sel_info.bg_par = {};
    used_ds('Ei400Br110dir001') = sel_info;
    %
    sel_info.cuts = cuts2fit400.cuts_dir001FFfit(2);
    sel_info.en_range = 50:10:200;
    sel_info.cut_dim = 2;
    sel_info.fname = 'EnFit_Ei400ref001off200_2D_constFFCut';
    sel_info.bg_par = {};
    used_ds('Ei400Br200dir001') = sel_info;

    %
    sel_info.cuts = cuts2fit400.cuts_dir001FFfit(1);
    sel_info.en_range = 50:10:160;
    sel_info.cut_dim = 1;
    sel_info.fname = 'EnFit_Ei400ref001off110_1D_constFFCut';
    sel_info.bg_par = {};
    used_ds('Ei400Br110dir001_1D') = sel_info;
    %
    sel_info.cuts = cuts2fit400.cuts_dir001FFfit(2);
    sel_info.en_range = 50:10:200;
    sel_info.cut_dim = 1;
    sel_info.fname = 'EnFit_Ei400ref001off200_1D_constFFCut';
    sel_info.bg_par = {};
    used_ds('Ei400Br200dir001_1D') = sel_info;
    used_ds400 = used_ds;
end

```

used_ds =

Map with properties:

```

    Count: 0
    KeyType: char
    ValueType: any

```

Show all accessible properties of Map

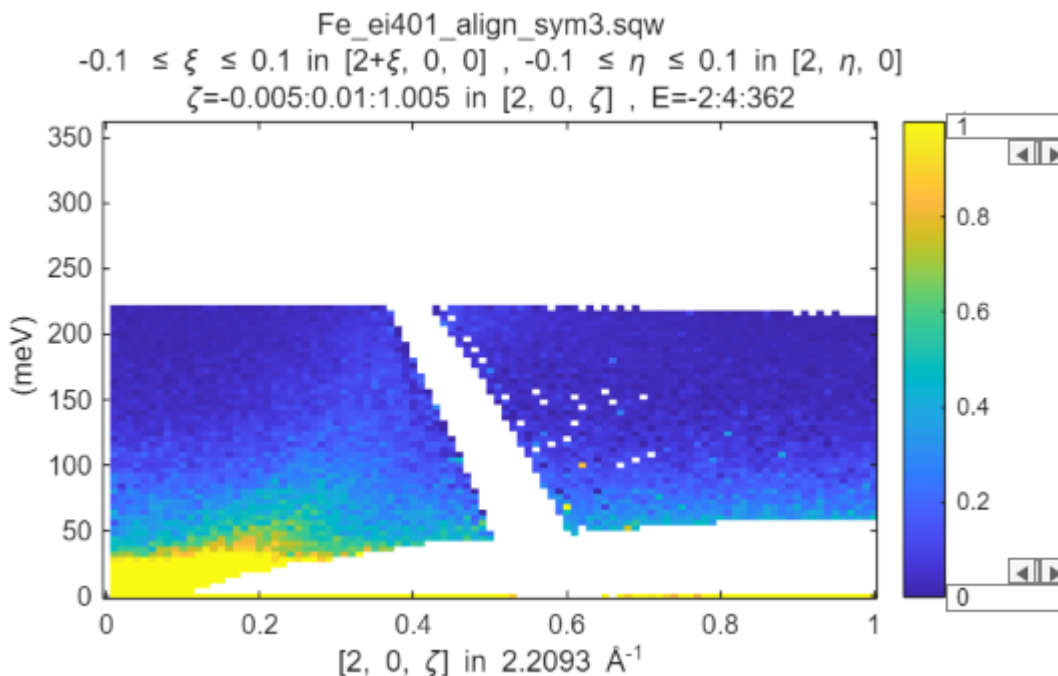
```

%cuts_list = cuts2fit400.cutsS_list; % all available cuts
%cuts_list = {cuts2fit400.cut_b200_200sel}; % cuts in low ff direction around
<200>
%cuts_list = cuts2fit400.cuts_dir001FFfit(1);
select= 'Fe400Br200dir001_1D';
sel_info = used_ds(select);

cuts_list = sel_info.cuts;

mi = maps_instrument(400,600,'S');
sample=IX_sample(true,[1,0,0],[0,1,0],'cuboid',[0.04,0.03,0.02]);
for i=1:numel(cuts_list)
    cuts_list{i} = cuts_list{i}.set_instrument(mi);
    cuts_list{i} = cuts_list{i}.set_sample(sample);
    plot(cuts_list{i}); lz 0 1; keep_figure;
end

```



```

dir_name = "GH";

dE_step = 4; %original energy transfer step data were binned to. No point in
going finer
half_dE = 5; % half width of data binning
%cut_en = 50:10:260;
%cut_en = 50:10:160; %cuts2fit400.cuts_dir001FFfit(1);
cut_en = sel_info.en_range; %cuts2fit400.cuts_dir001FFfit(2);
cut_dimensions = sel_info.cut_dim;

%cl= {cuts_list{3}};
%cl = {w2_test_cut};
% if isfield(cuts2fit400,'bg_par_010off000')

```

```

%     if numel(cuts_list) == 3
%         bg_par =
{cuts2fit400.bg_par_010off000.p,cuts2fit400.bg_par_010off100.p,cuts2fit400.bg_par
_010off200.p};
%     else
%         bg_par = {cuts2fit400.bg_par_010off100.p};
%     end
%     cuts_list{end+1} = bg_par;
% end
bg_par = sel_info.bg_par;
cuts_list{end+1} = bg_par;
fbase_name = sel_info.fname;

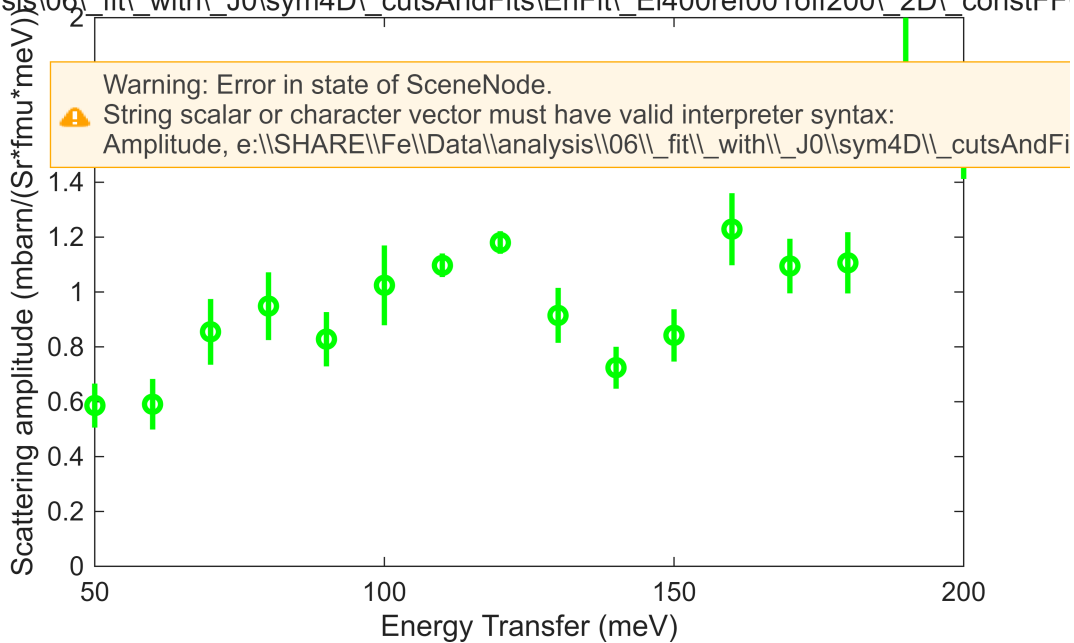
```

```

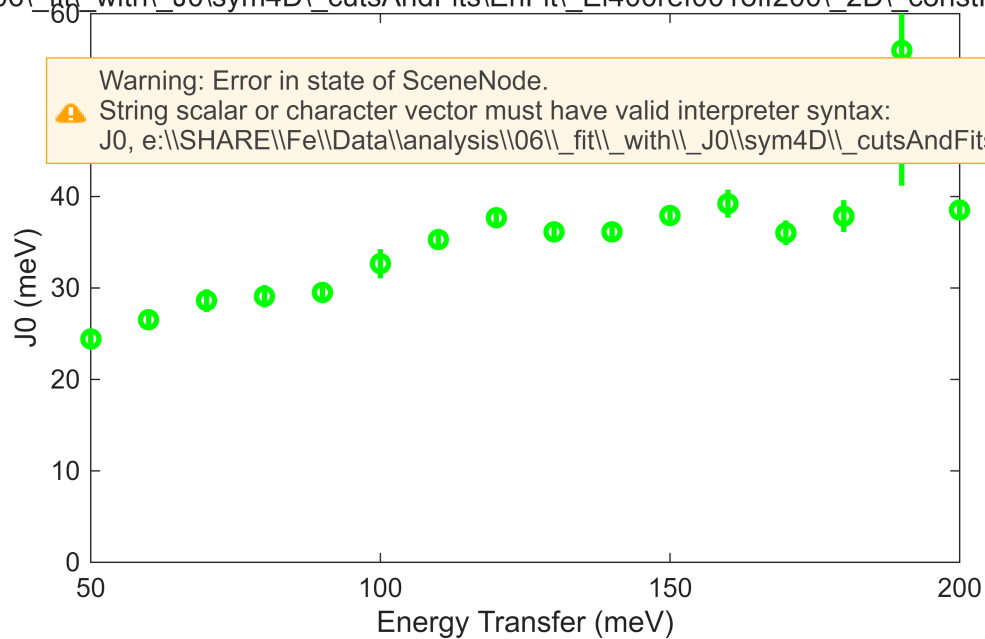
[fit_res_400rec,res_name] = fit_multicuts_along_direction(...
    cuts_list,fbase_name,dir_name,cut_dimensions,cut_en,dE_step,half_dE,true);

```

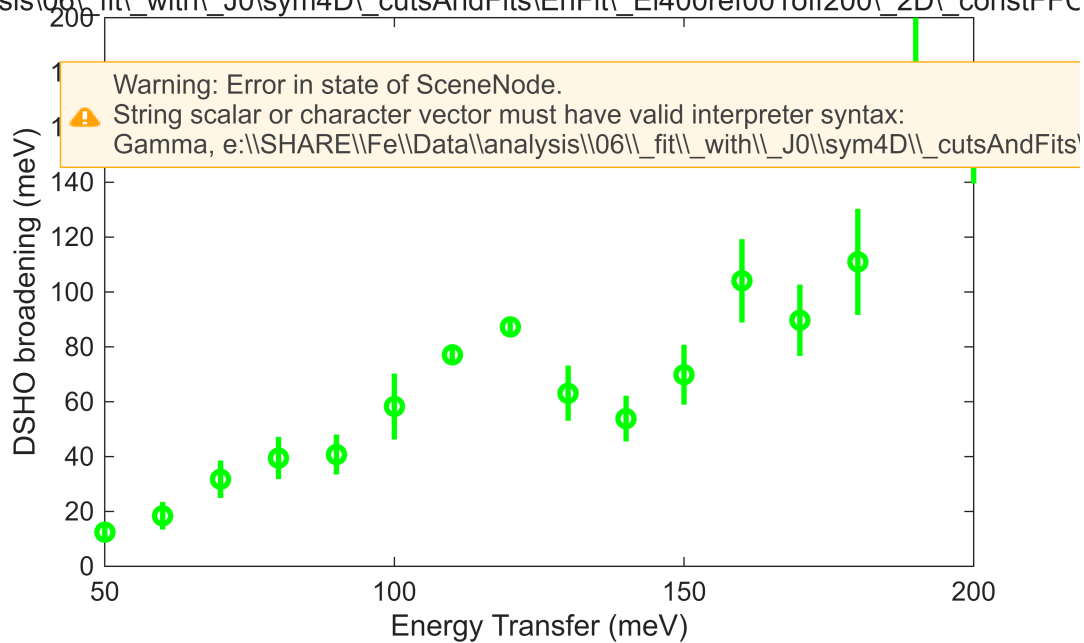
analysis\06\ fit_with_J0\sym4D_cutsAndFits\EnFit_Ei400ref001off200_2D_constFF



'sis\06\ _fit\ _with_J0\sym4D_cutsAndFits\EnFit_Ei400ref001off200_2D_constFFCut\



alysis\06\ _fit\ _with_J0\sym4D_cutsAndFits\EnFit_Ei400ref001off200_2D_constFFC

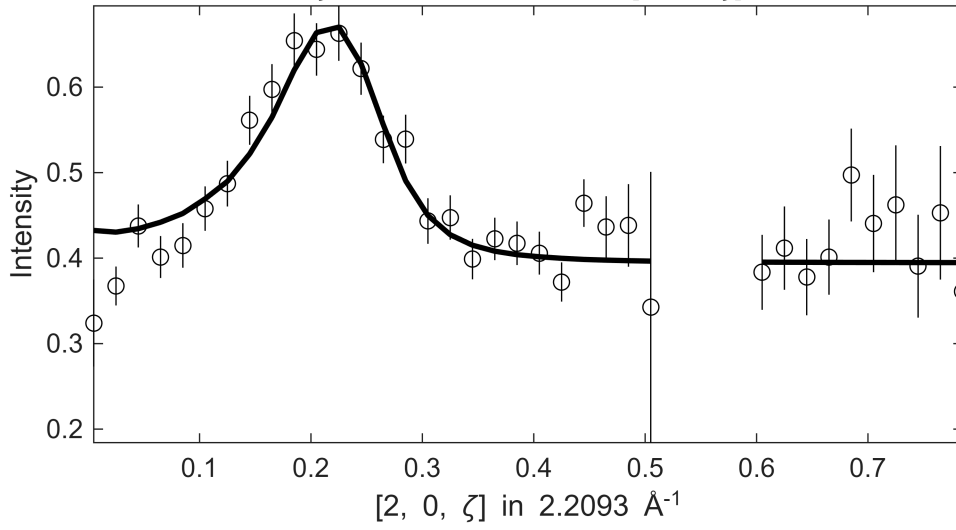


 **** En = 50±5

Fe_ei401_align_sym3.sqw

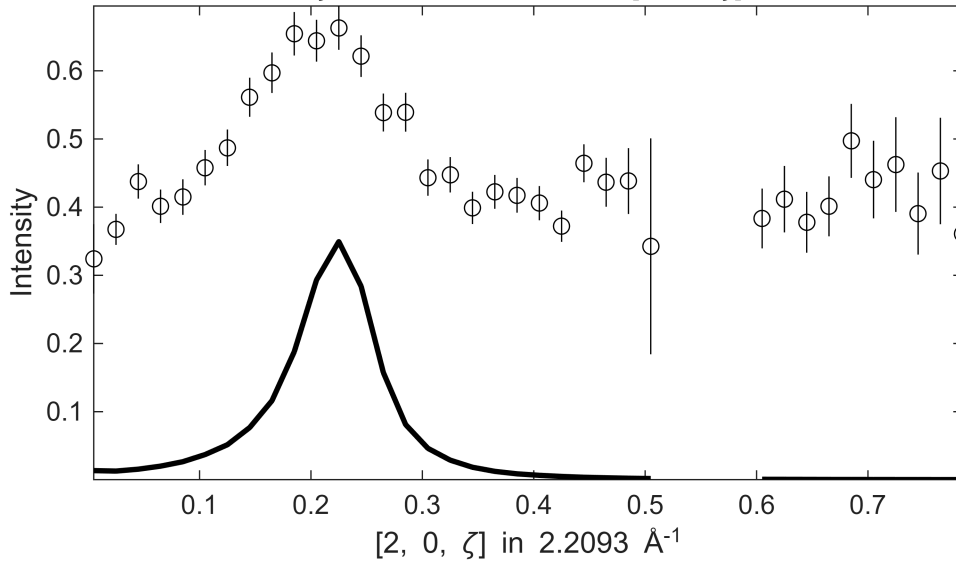
S=0.61±0.034; J0= 25±0.75; gamma=13.7434±0.688257

-0.1 ≤ ξ ≤ 0.1 in [2+ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, η, 0] , 45 ≤ E ≤ 55
ζ=-0.005:0.02:1.015 in [2, 0, ζ]



Fe_ei401_align_sym3.sqw

-0.1 ≤ ξ ≤ 0.1 in [2+ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, η, 0] , 45 ≤ E ≤ 55
ζ=-0.005:0.02:1.015 in [2, 0, ζ]

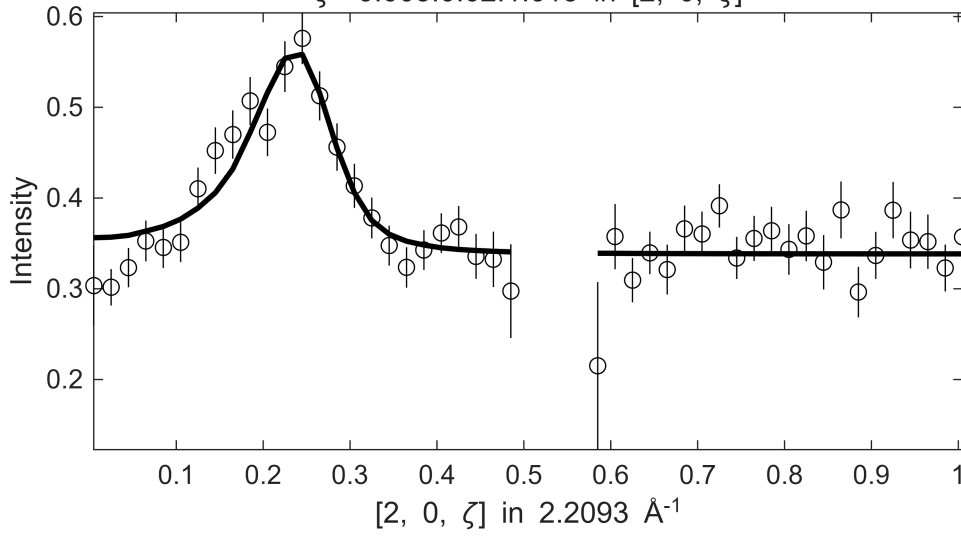


**** En = 60±5

Fe_ei401_align_sym3.sqw

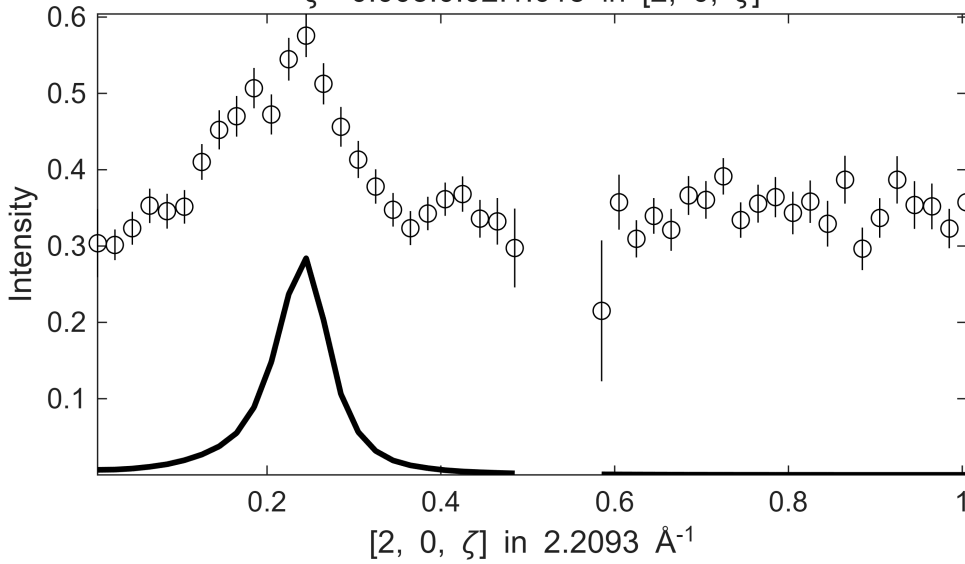
S=0.51±0.074; J0= 26±0.86; gamma=13.9969±4.09096

-0.1 ≤ ξ ≤ 0.1 in [2+ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, η, 0] , 55 ≤ E ≤ 65
ζ=-0.005:0.02:1.015 in [2, 0, ζ]



Fe_ei401_align_sym3.sqw

-0.1 ≤ ξ ≤ 0.1 in [2+ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, η, 0] , 55 ≤ E ≤ 6
ζ=-0.005:0.02:1.015 in [2, 0, ζ]



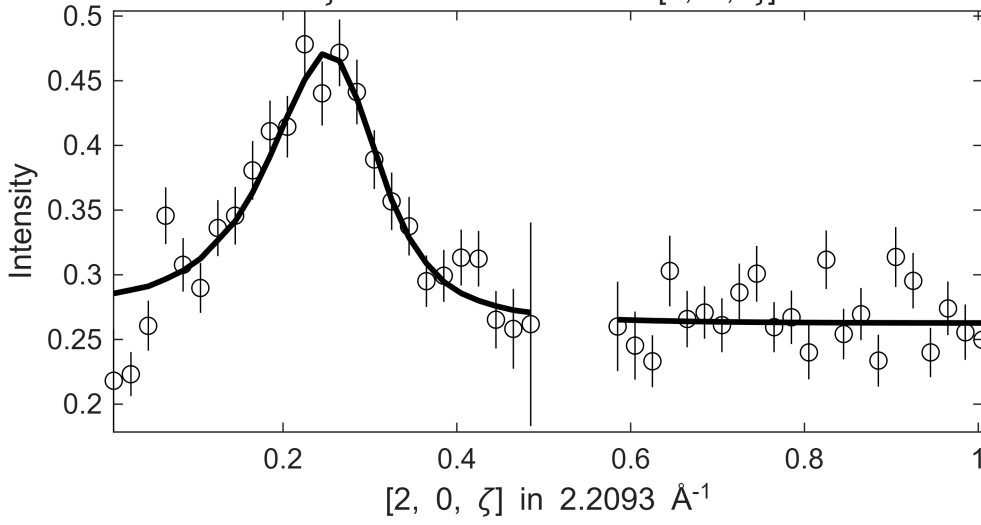
**** En = 70±5

Fe_ei401_align_sym3.sqw

S=0.86±0.045; J0= 29±0.88; gamma=33.9505±1.50365

-0.1 ≤ ξ ≤ 0.1 in [2+ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, η, 0] , 65 ≤ E ≤ 7

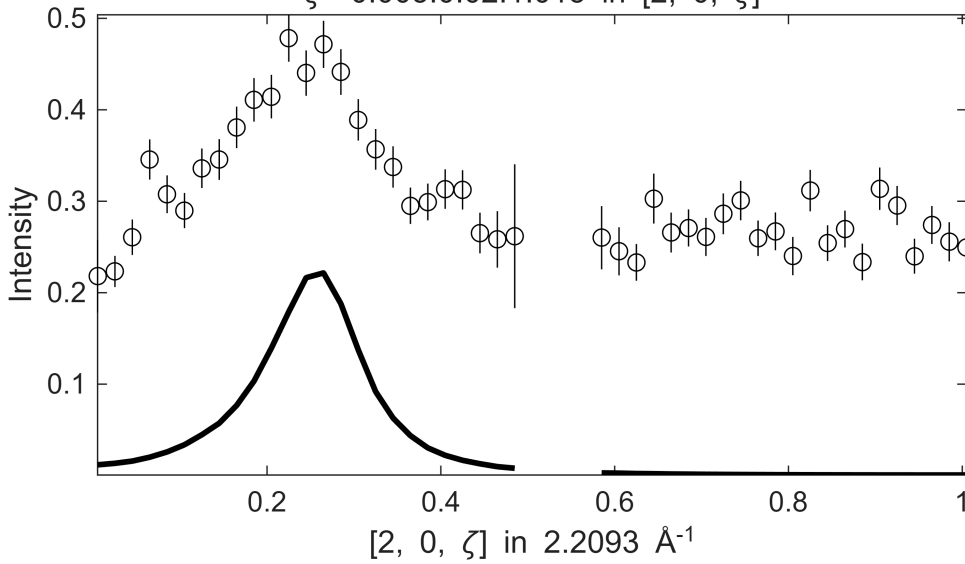
ζ=-0.005:0.02:1.015 in [2, 0, ζ]



Fe_ei401_align_sym3.sqw

-0.1 ≤ ξ ≤ 0.1 in [2+ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, η, 0] , 65 ≤ E ≤ 7

ζ=-0.005:0.02:1.015 in [2, 0, ζ]



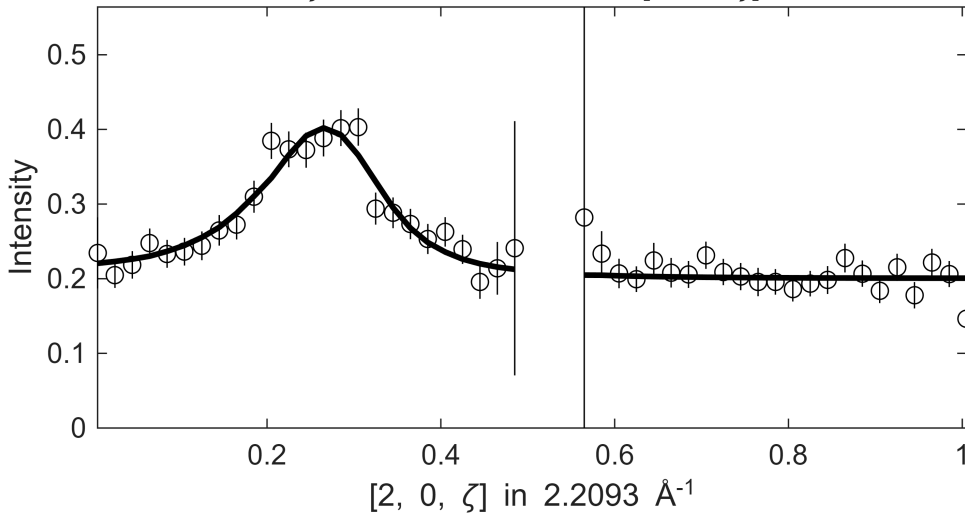
**** En = 80±5

Fe_ei401_align_sym3.sqw

S= 1 ± 0.037 ; J0= 30 ± 0.68 ; gamma= 42.8788 ± 1.28076

$-0.1 \leq \xi \leq 0.1$ in $[2+\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, \eta, 0]$, $75 \leq E \leq 8$

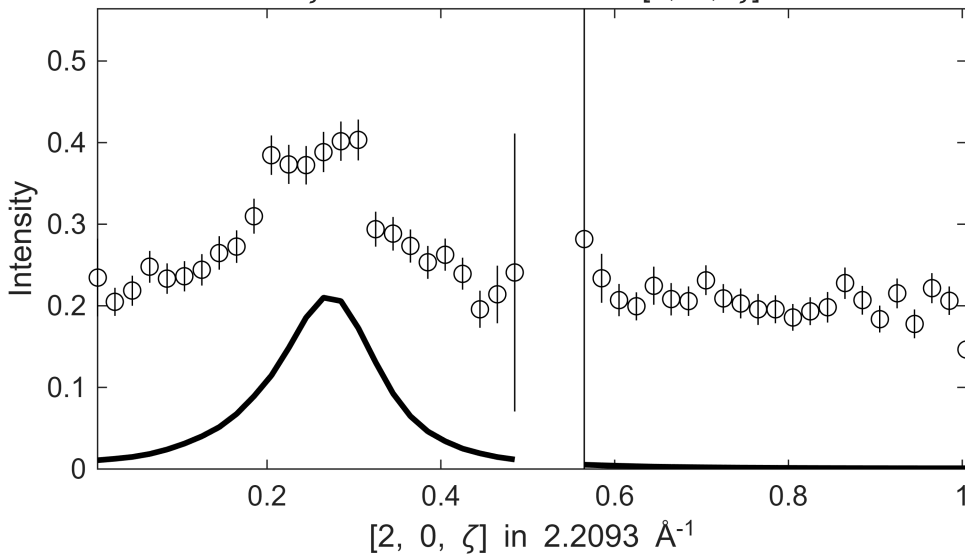
$\zeta = -0.005:0.02:1.015$ in $[2, 0, \zeta]$



Fe_ei401_align_sym3.sqw

$-0.1 \leq \xi \leq 0.1$ in $[2+\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, \eta, 0]$, $75 \leq E \leq 8$

$\zeta = -0.005:0.02:1.015$ in $[2, 0, \zeta]$



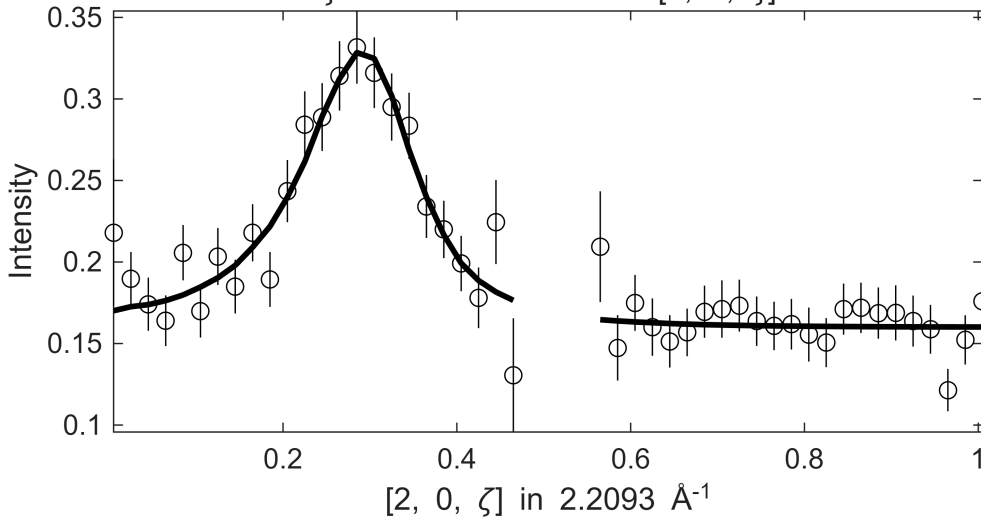
**** En = 90 ± 5

Fe_ei401_align_sym3.sqw

S=0.78±0.031; J0= 29±0.62; gamma=37.9373±1.23645

-0.1 ≤ ξ ≤ 0.1 in [2+ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, η, 0] , 85 ≤ E ≤ 9

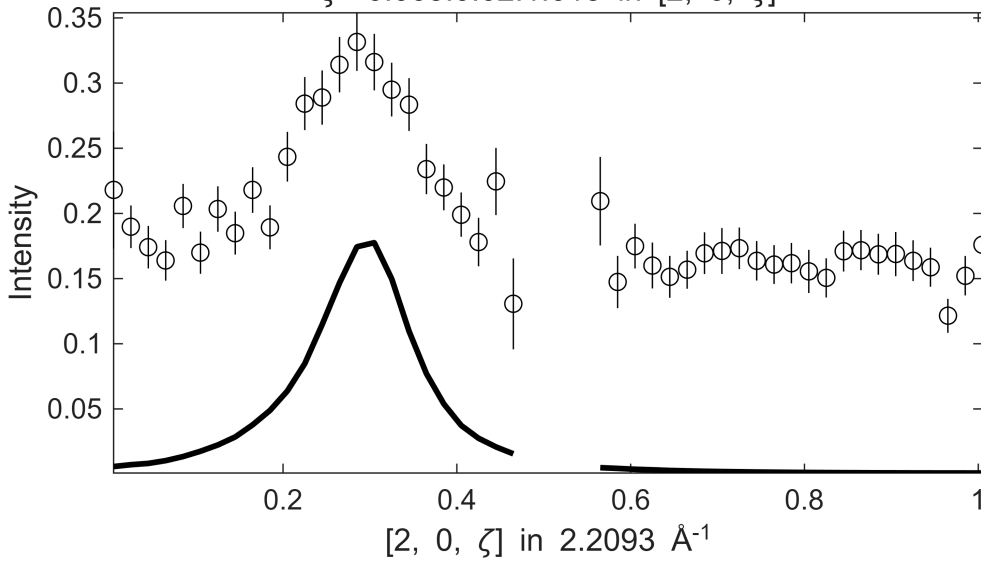
ζ=-0.005:0.02:1.015 in [2, 0, ζ]



Fe_ei401_align_sym3.sqw

-0.1 ≤ ξ ≤ 0.1 in [2+ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, η, 0] , 85 ≤ E ≤ 9

ζ=-0.005:0.02:1.015 in [2, 0, ζ]



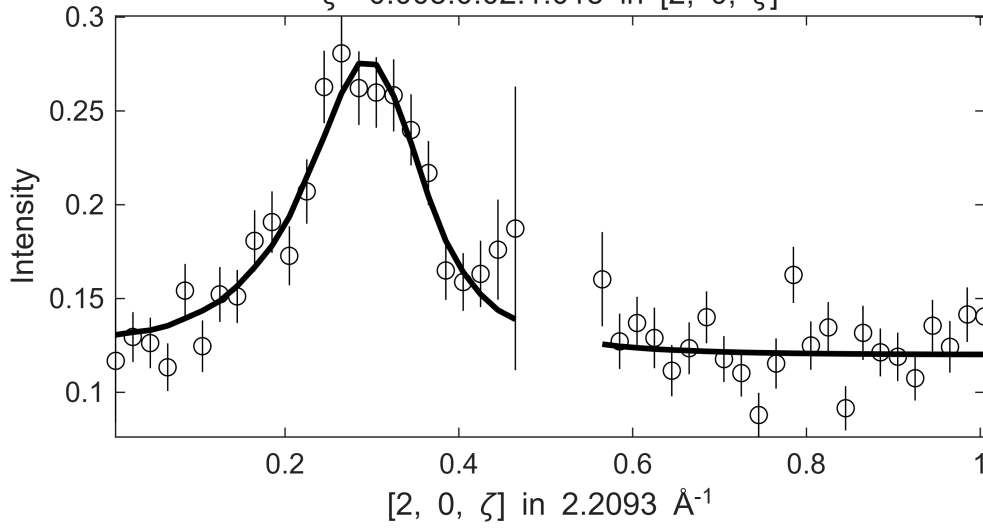
**** En = 100±5

Fe_ei401_align_sym3.sqw

S=0.86±0.13; J0= 31± 1.4; gamma=46.6312±10.0811

-0.1 ≤ ξ ≤ 0.1 in [2+ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, η, 0] , 95 ≤ E ≤ 100

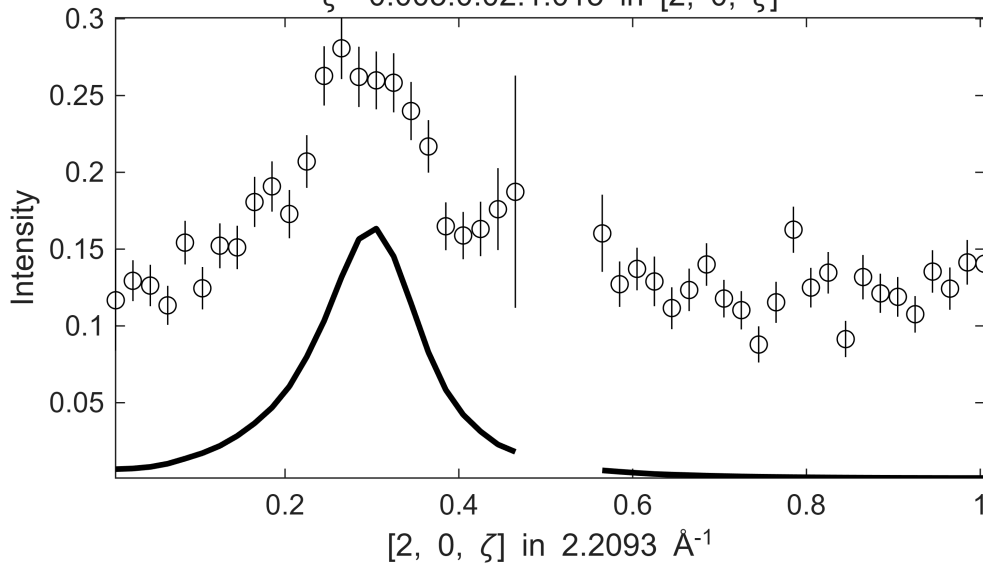
ζ=-0.005:0.02:1.015 in [2, 0, ζ]



Fe_ei401_align_sym3.sqw

-0.1 ≤ ξ ≤ 0.1 in [2+ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, η, 0] , 95 ≤ E ≤ 100

ζ=-0.005:0.02:1.015 in [2, 0, ζ]



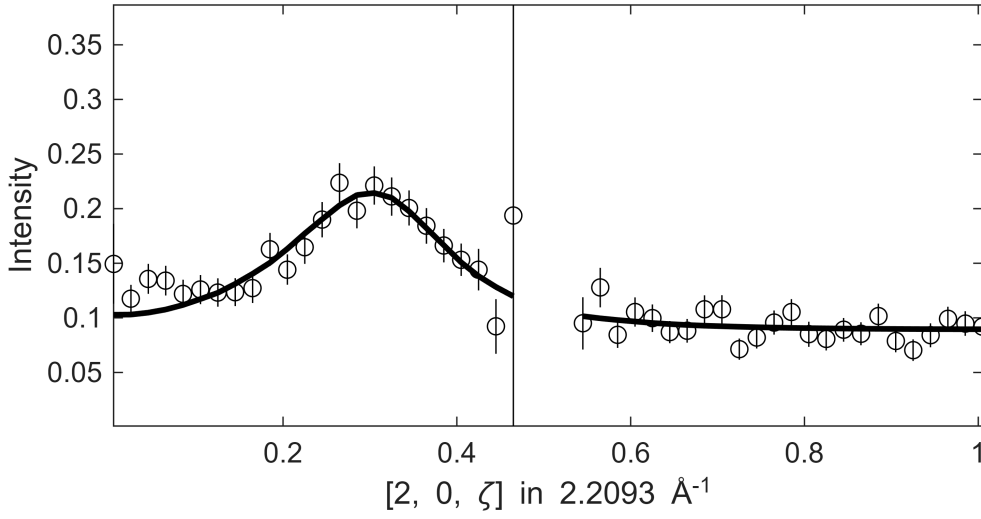
**** En = 110±5

Fe_ei401_align_sym3.sqw

S= 1.4±0.06; J0= 39± 1.1; gamma=108.271±3.36089

-0.1 ≤ ξ ≤ 0.1 in [2+ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, η, 0] , 105 ≤ E ≤ 1

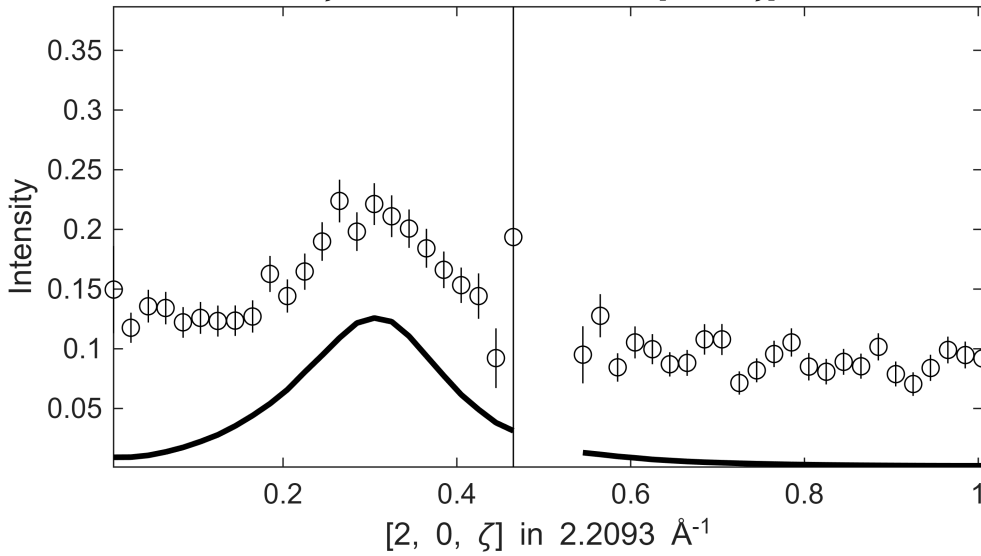
ζ=-0.005:0.02:1.015 in [2, 0, ζ]



Fe_ei401_align_sym3.sqw

-0.1 ≤ ξ ≤ 0.1 in [2+ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, η, 0] , 105 ≤ E ≤ 1

ζ=-0.005:0.02:1.015 in [2, 0, ζ]



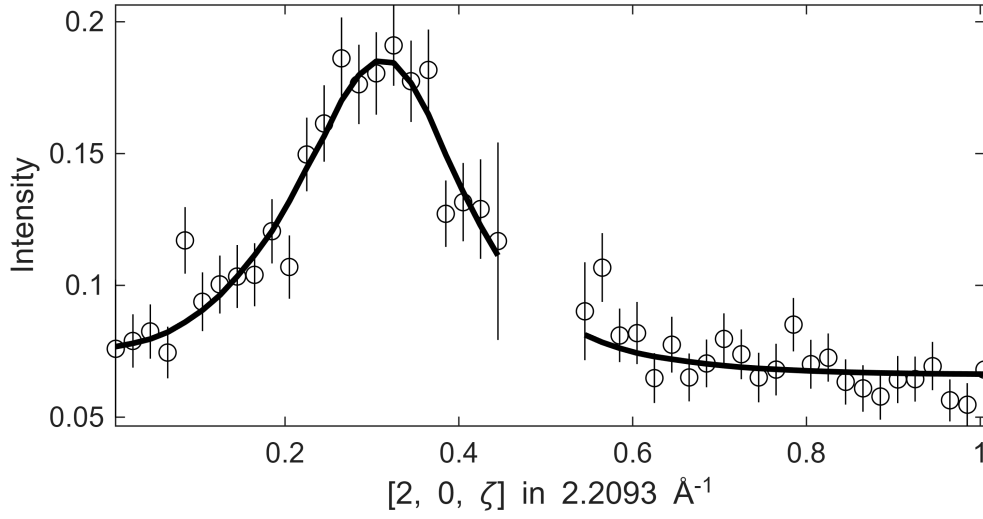
**** En = 120±5

Fe_ei401_align_sym3.sqw

S= 1.5±0.06; J0= 41± 1.1; gamma=121.766±3.13656

-0.1 ≤ ξ ≤ 0.1 in [2+ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, η, 0] , 115 ≤ E ≤ 1

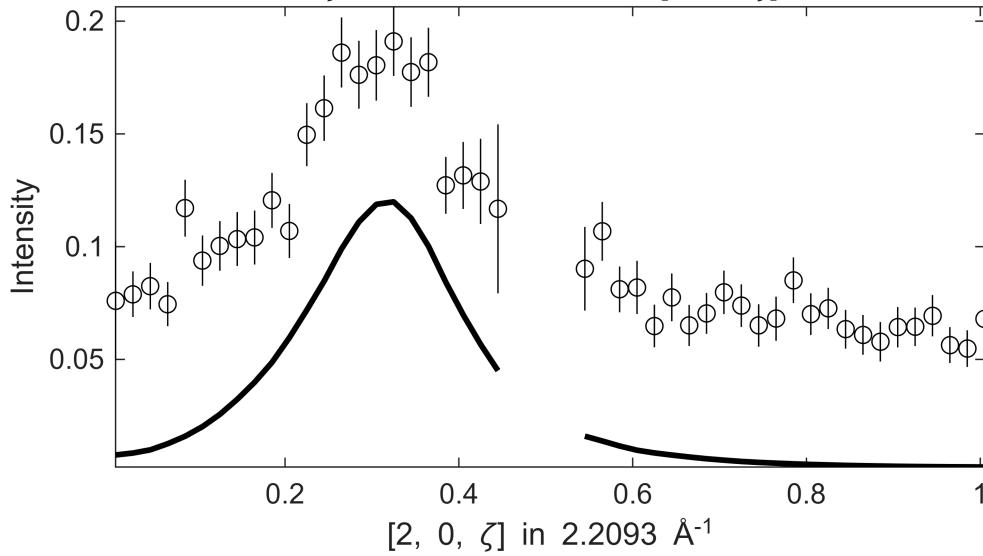
ζ=-0.005:0.02:1.015 in [2, 0, ζ]



Fe_ei401_align_sym3.sqw

-0.1 ≤ ξ ≤ 0.1 in [2+ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, η, 0] , 115 ≤ E ≤ 1

ζ=-0.005:0.02:1.015 in [2, 0, ζ]



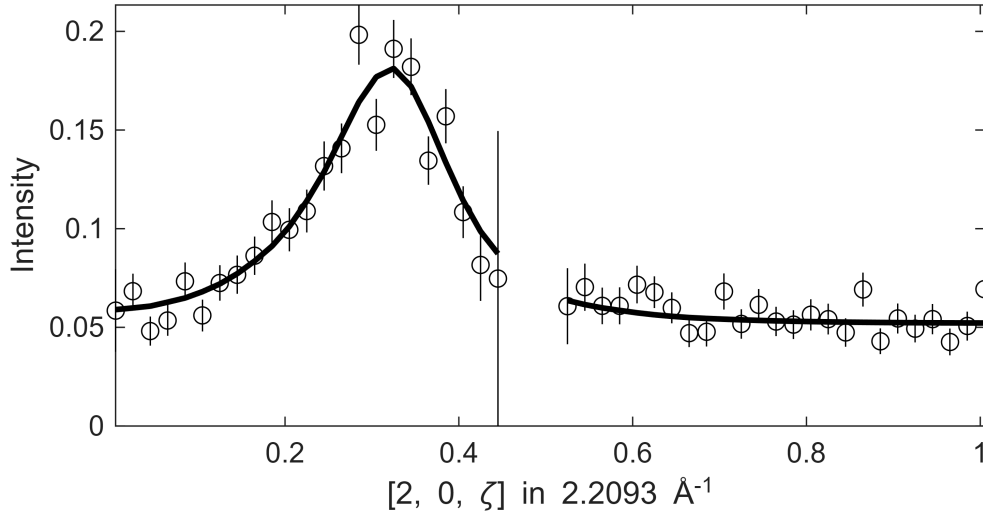
**** En = 130±5

Fe_ei401_align_sym3.sqw

S=0.97±0.12; J0= 36± 1.3; gamma=65.439±11.8935

-0.1 ≤ ξ ≤ 0.1 in [2+ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, η, 0] , 125 ≤ E ≤ 1

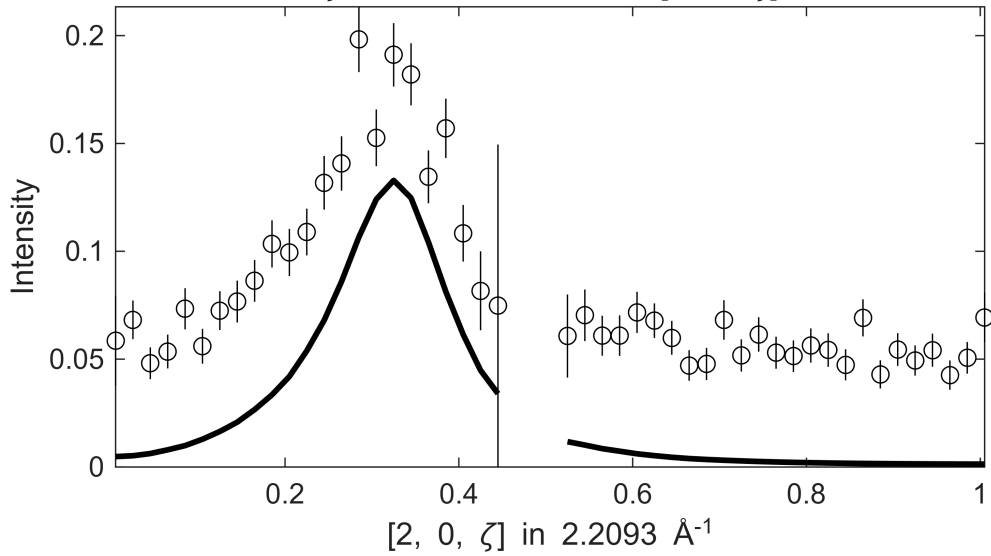
ζ=-0.005:0.02:1.015 in [2, 0, ζ]



Fe_ei401_align_sym3.sqw

-0.1 ≤ ξ ≤ 0.1 in [2+ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, η, 0] , 125 ≤ E ≤ 1

ζ=-0.005:0.02:1.015 in [2, 0, ζ]



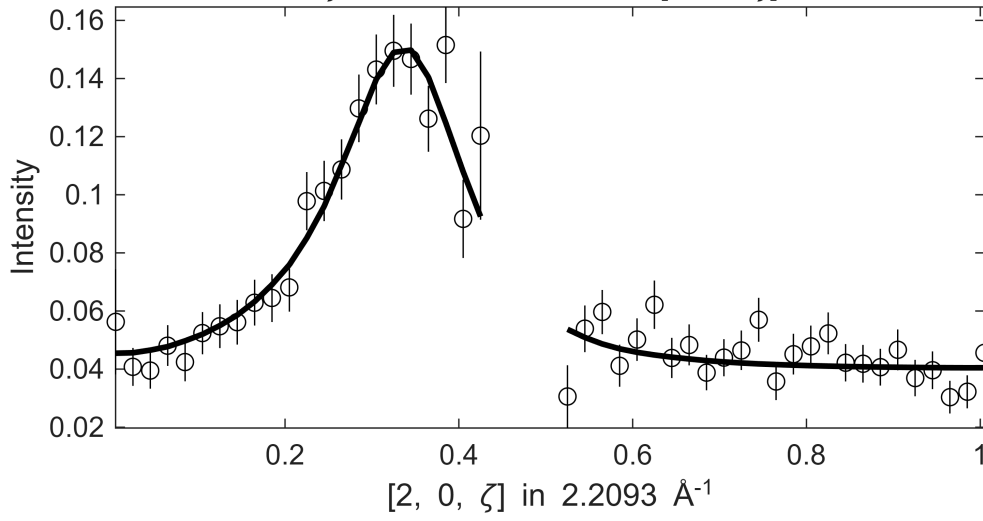
**** En = 140±5

Fe_ei401_align_sym3.sqw

S=0.87±0.11; J0= 36± 1.2; gamma=68.3295±12.2457

-0.1 ≤ ξ ≤ 0.1 in [2+ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, η, 0] , 135 ≤ E ≤ 1

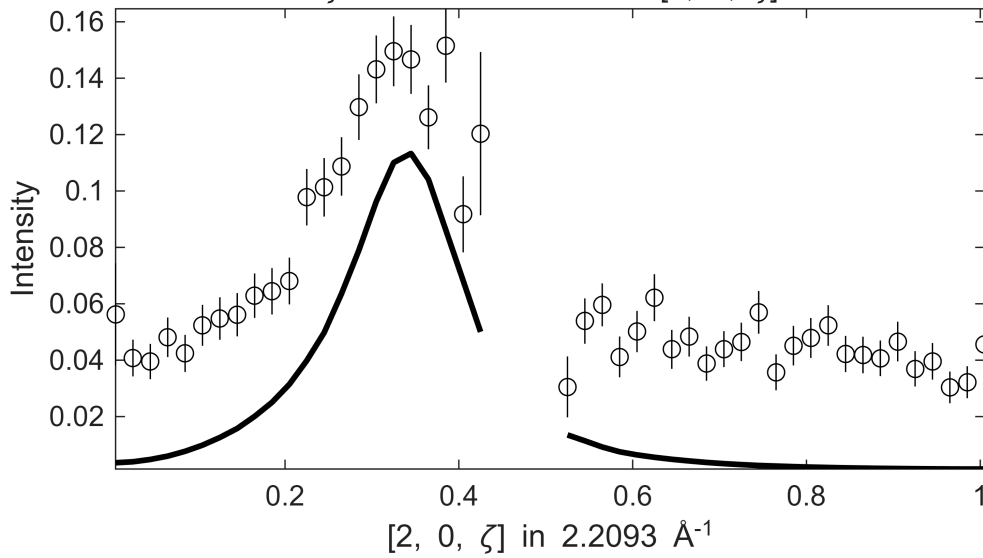
ζ=-0.005:0.02:1.015 in [2, 0, ζ]



Fe_ei401_align_sym3.sqw

-0.1 ≤ ξ ≤ 0.1 in [2+ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, η, 0] , 135 ≤ E ≤ 1

ζ=-0.005:0.02:1.015 in [2, 0, ζ]



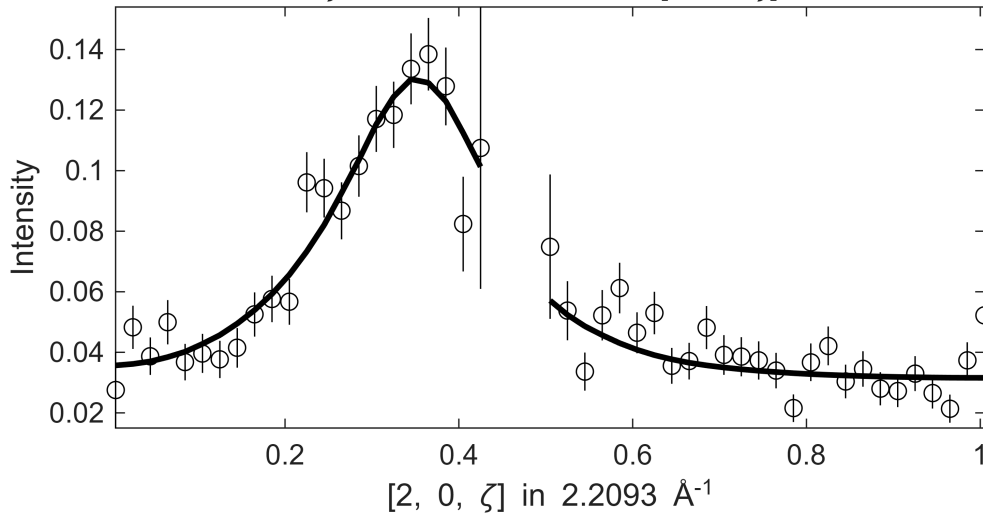
**** En = 150±5

Fe_ei401_align_sym3.sqw

S= 1.1± 0.2; J0= 37± 2.3; gamma=98.6148±24.1982

-0.1 ≤ ξ ≤ 0.1 in [2+ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, η, 0] , 145 ≤ E ≤ 1

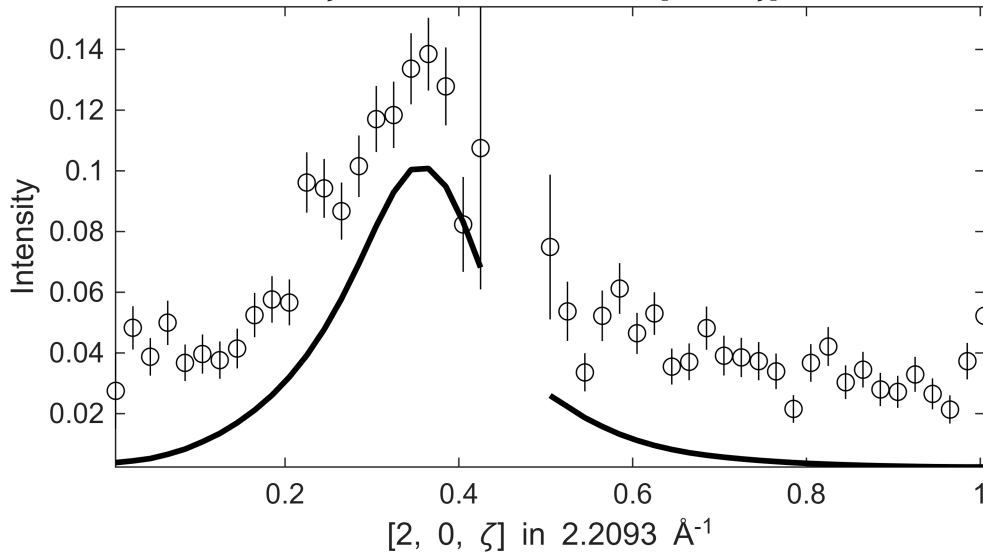
ζ=-0.005:0.02:1.015 in [2, 0, ζ]



Fe_ei401_align_sym3.sqw

-0.1 ≤ ξ ≤ 0.1 in [2+ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, η, 0] , 145 ≤ E ≤ 1

ζ=-0.005:0.02:1.015 in [2, 0, ζ]



**** En = 160±5

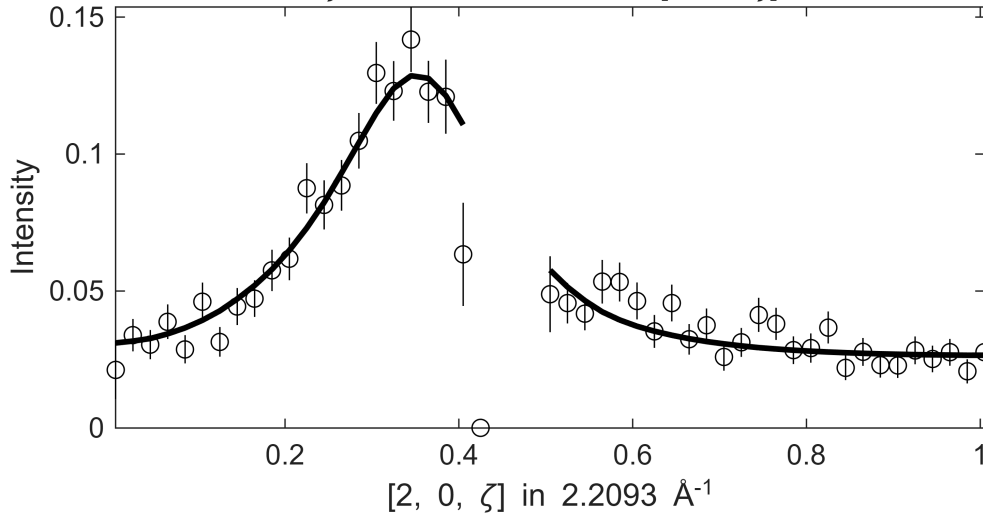
Warning: Dataset:1 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 1.9608

Fe_ei401_align_sym3.sqw

S= 1.4±0.22; J0= 42± 2.6; gamma=123.842±26.8504

-0.1 ≤ ξ ≤ 0.1 in [2+ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, η, 0] , 155 ≤ E ≤ 1

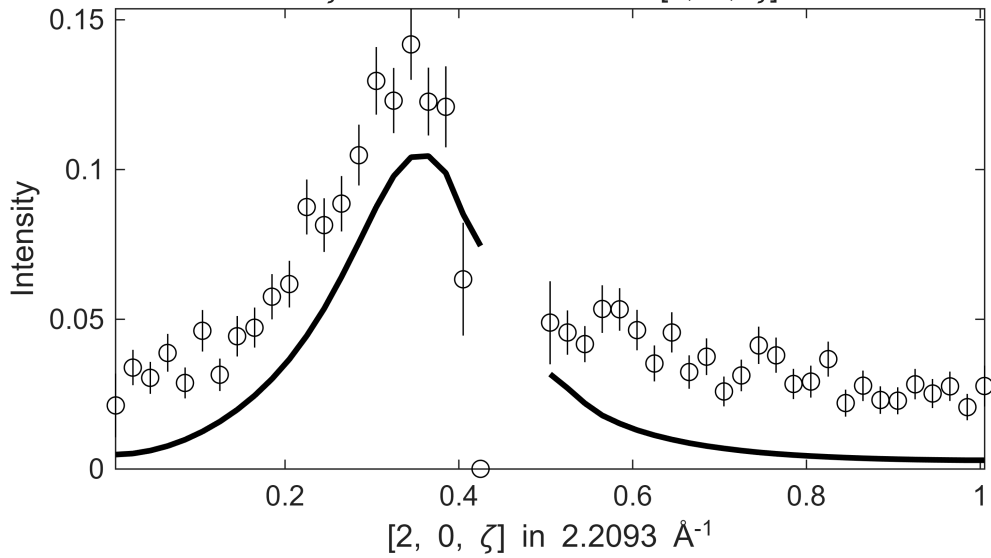
ζ=-0.005:0.02:1.015 in [2, 0, ζ]



Fe_ei401_align_sym3.sqw

-0.1 ≤ ξ ≤ 0.1 in [2+ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, η, 0] , 155 ≤ E ≤ 1

ζ=-0.005:0.02:1.015 in [2, 0, ζ]



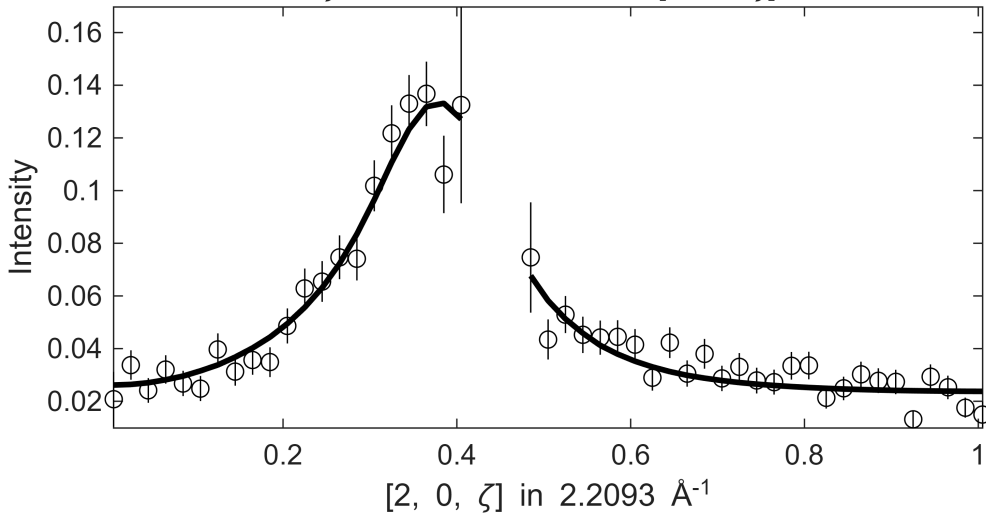
**** En = 170±5

Fe_ei401_align_sym3.sqw

S= 1.1±0.063; J0= 36±0.83; gamma=84.9098±3.33859

-0.1 ≤ ξ ≤ 0.1 in [2+ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, η, 0] , 165 ≤ E ≤ 1

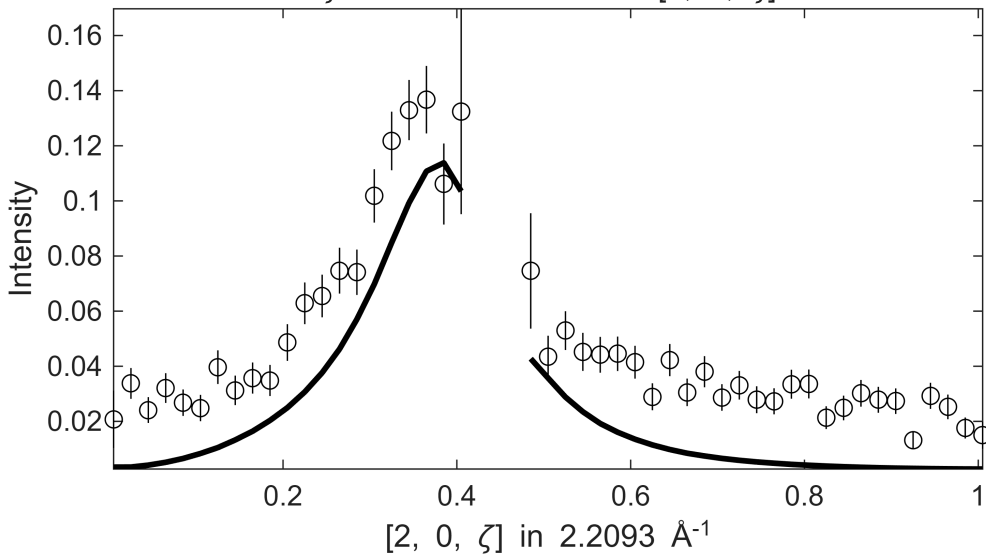
ζ=-0.005:0.02:1.015 in [2, 0, ζ]



Fe_ei401_align_sym3.sqw

-0.1 ≤ ξ ≤ 0.1 in [2+ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, η, 0] , 165 ≤ E ≤ 1

ζ=-0.005:0.02:1.015 in [2, 0, ζ]



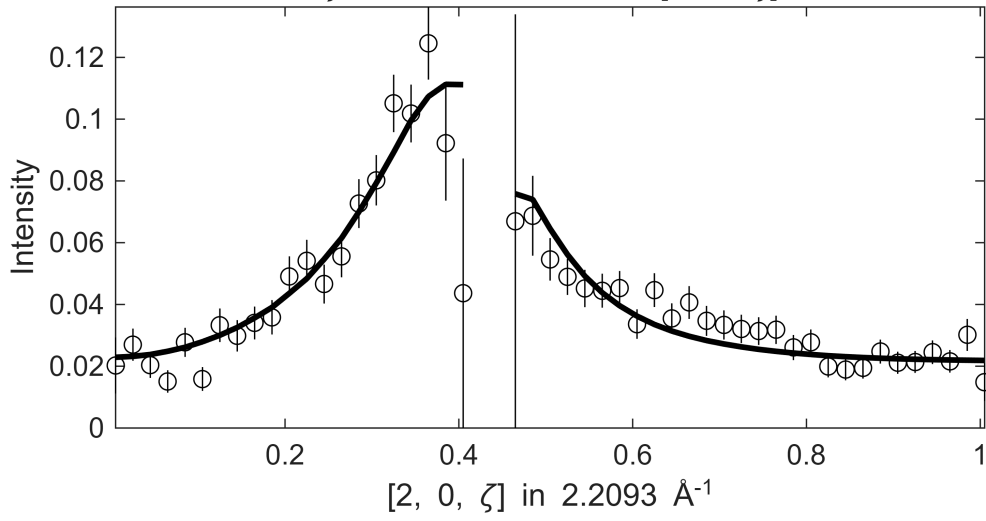
**** En = 180±5

Fe_ei401_align_sym3.sqw

S= 1.1±0.15; J0= 37± 2.3; gamma=107.986±25.6534

-0.1 ≤ ξ ≤ 0.1 in [2+ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, η, 0] , 175 ≤ E ≤ 1

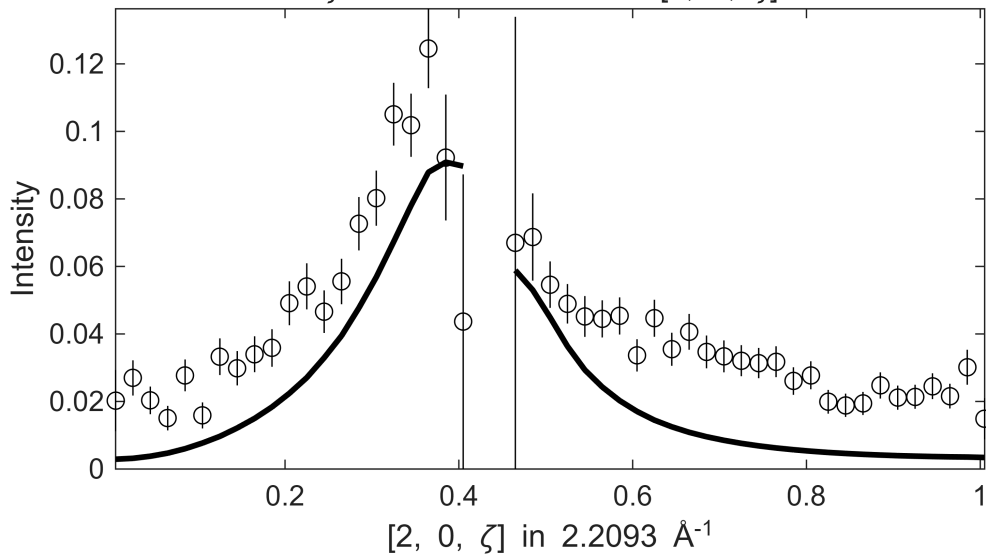
ζ=-0.005:0.02:1.015 in [2, 0, ζ]



Fe_ei401_align_sym3.sqw

-0.1 ≤ ξ ≤ 0.1 in [2+ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, η, 0] , 175 ≤ E ≤ 1

ζ=-0.005:0.02:1.015 in [2, 0, ζ]



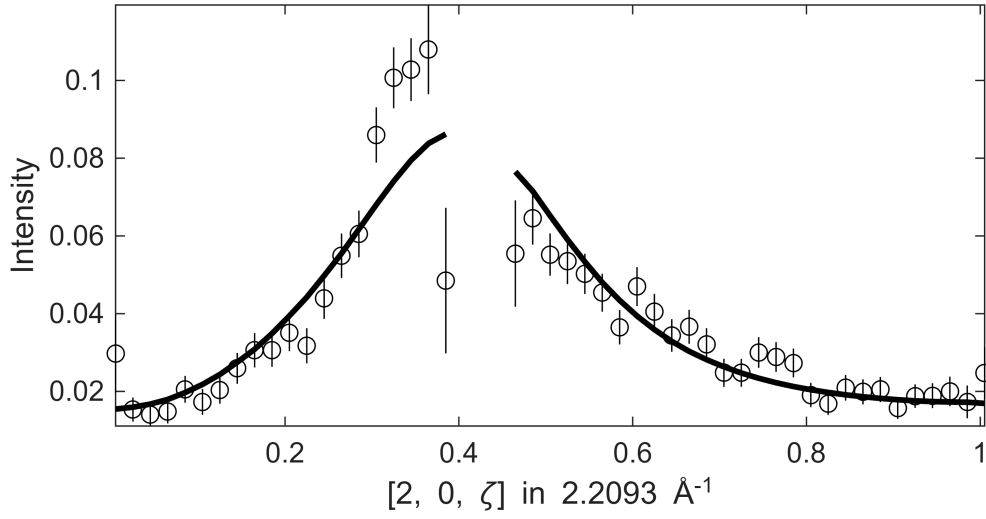
**** En = 190±5

Fe_ei401_align_sym3.sqw

S= 2 ± 0.79 ; J0= 49 ± 12 ; gamma= 288.285 ± 176.774

$-0.1 \leq \xi \leq 0.1$ in $[2+\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, \eta, 0]$, $185 \leq E \leq 1$

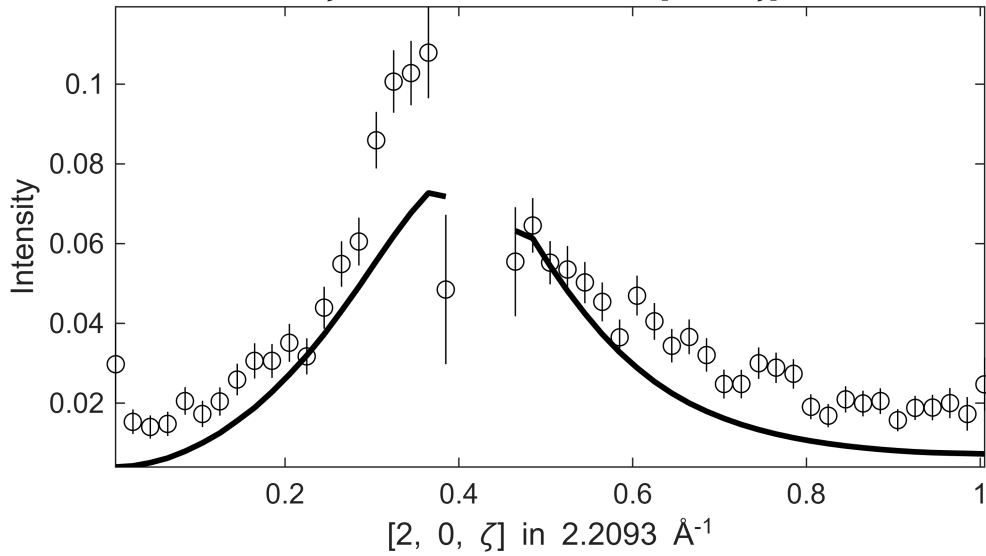
$\zeta = -0.005:0.02:1.015$ in $[2, 0, \zeta]$



Fe_ei401_align_sym3.sqw

$-0.1 \leq \xi \leq 0.1$ in $[2+\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, \eta, 0]$, $185 \leq E \leq 1$

$\zeta = -0.005:0.02:1.015$ in $[2, 0, \zeta]$



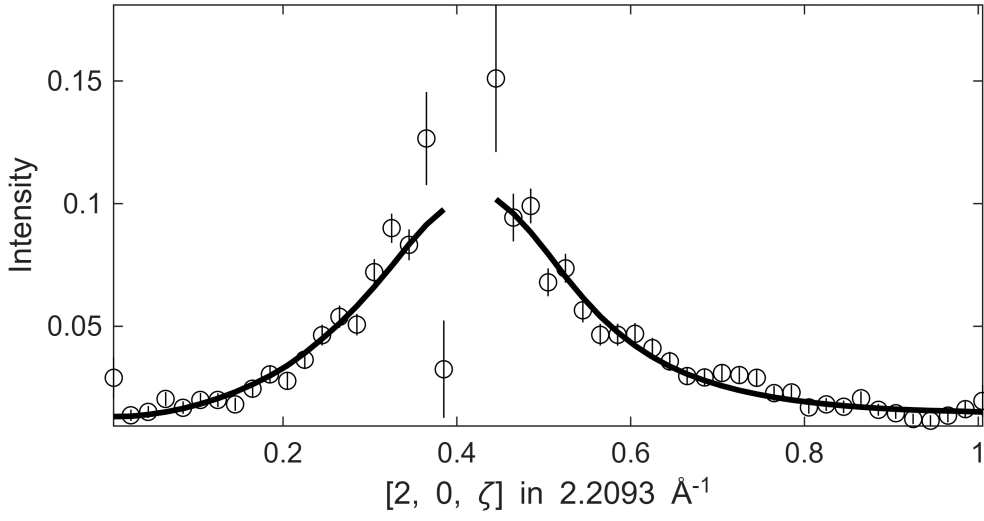
**** En = 200 ± 5

Fe_ei401_align_sym3.sqw

S= 1.5 ± 0.11 ; J0= 38 ± 0.8 ; gamma= 152.139 ± 12.7232

$-0.1 \leq \xi \leq 0.1$ in $[2+\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, \eta, 0]$, $195 \leq E \leq 2$

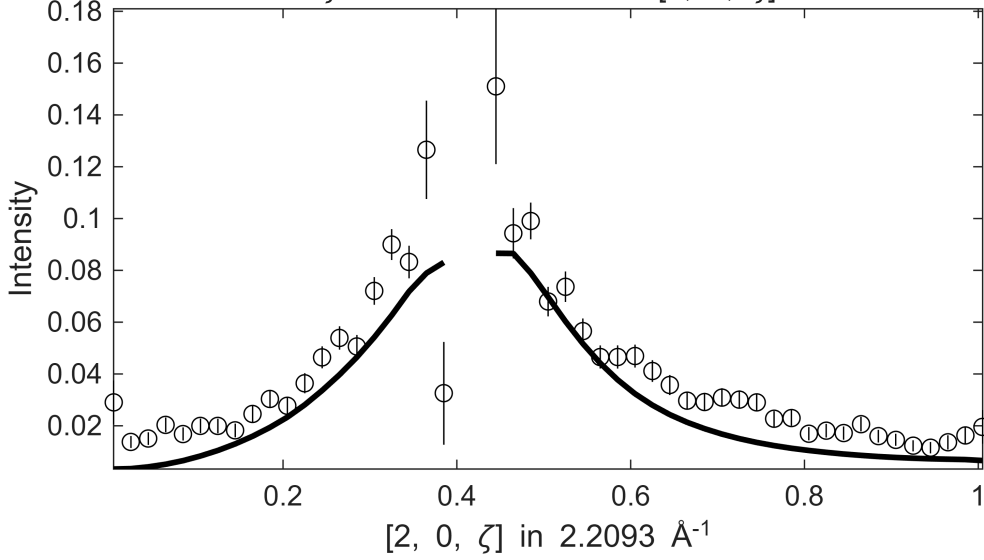
$\zeta = -0.005:0.02:1.015$ in $[2, 0, \zeta]$

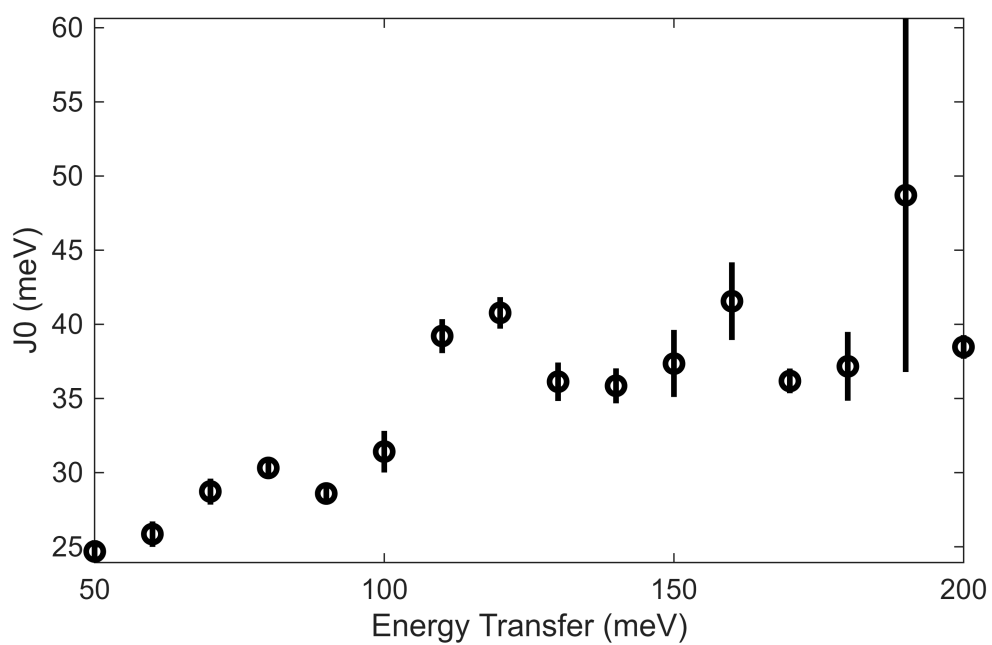
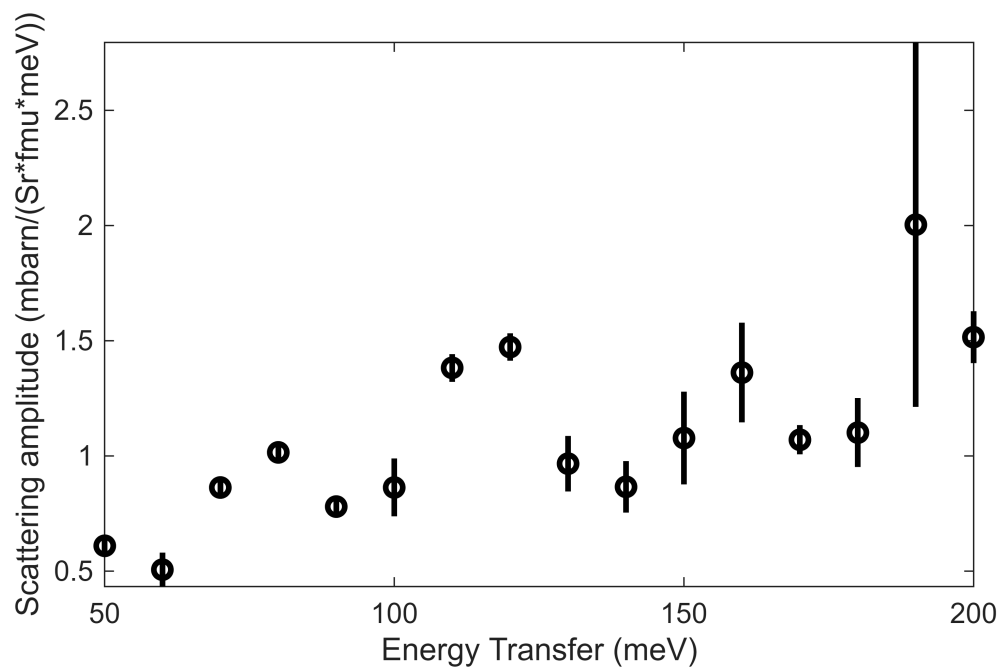


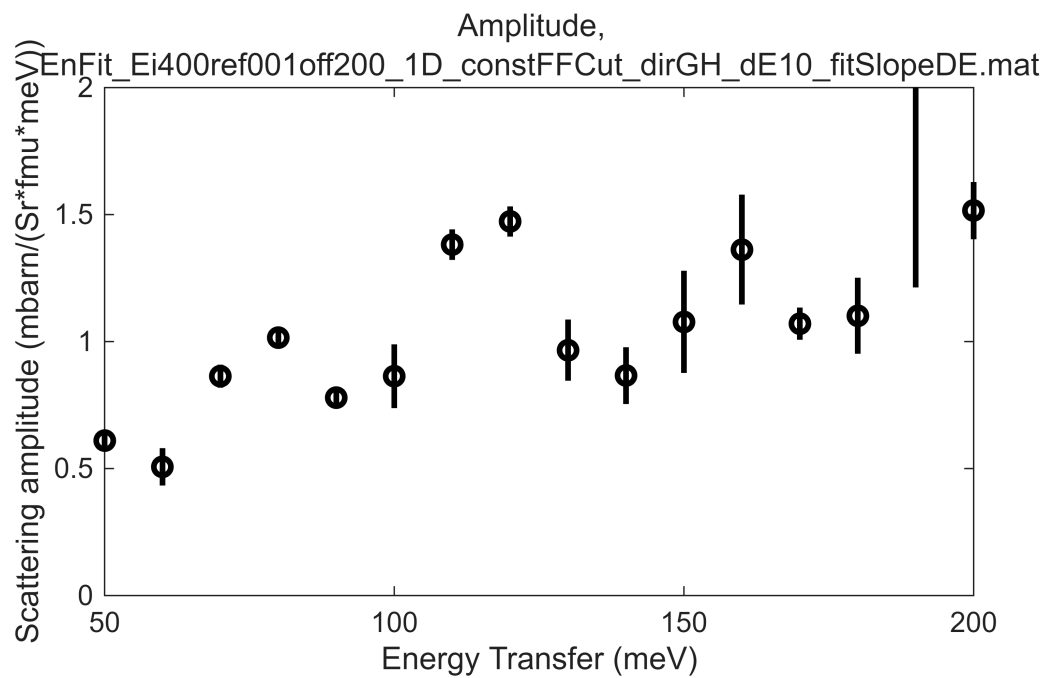
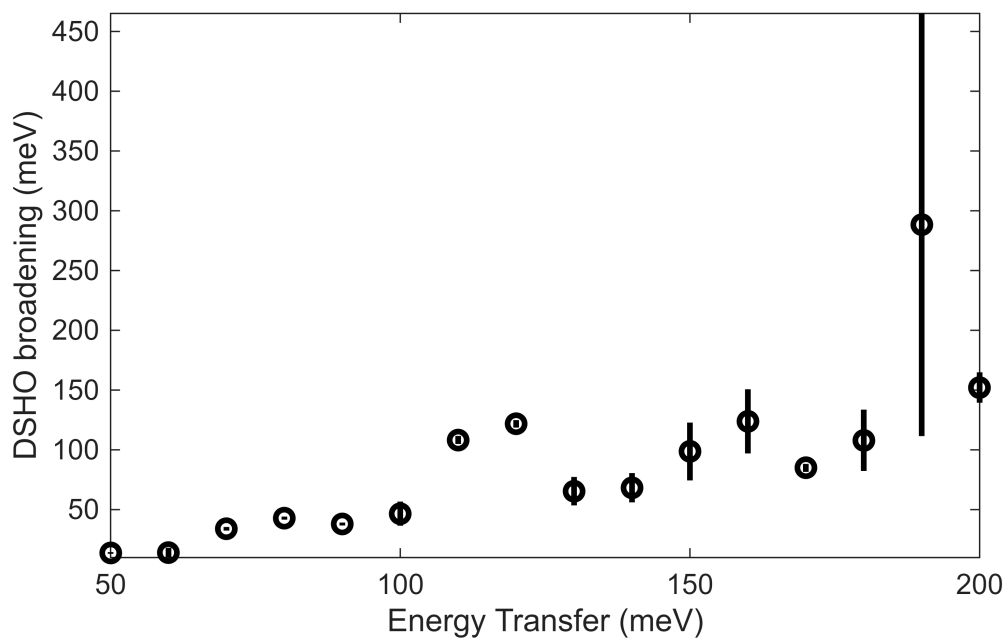
Fe_ei401_align_sym3.sqw

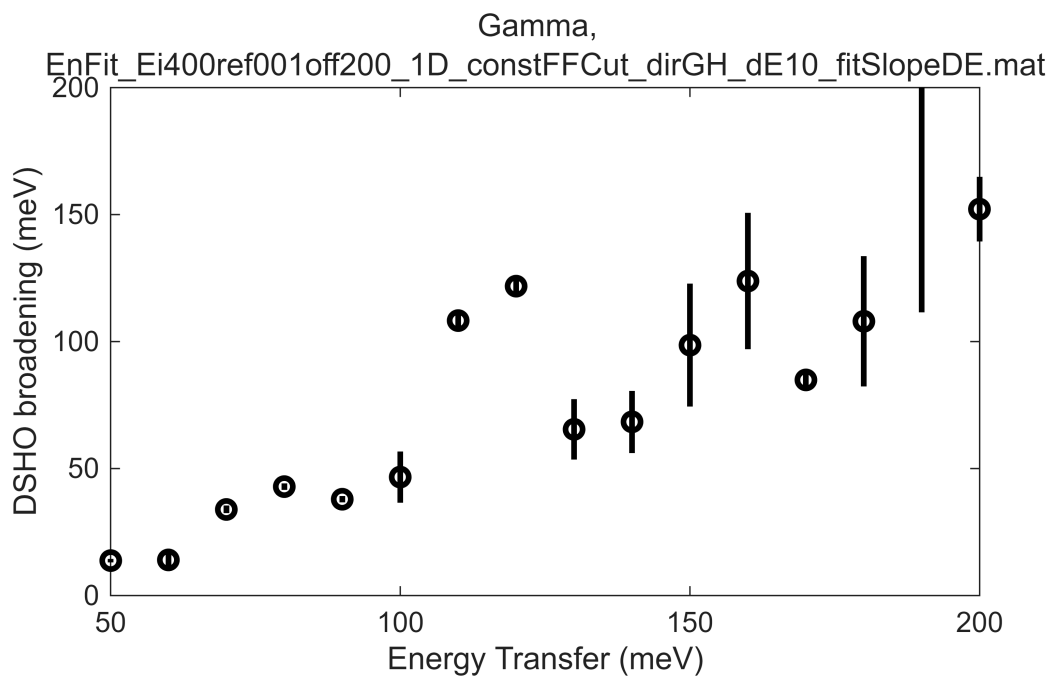
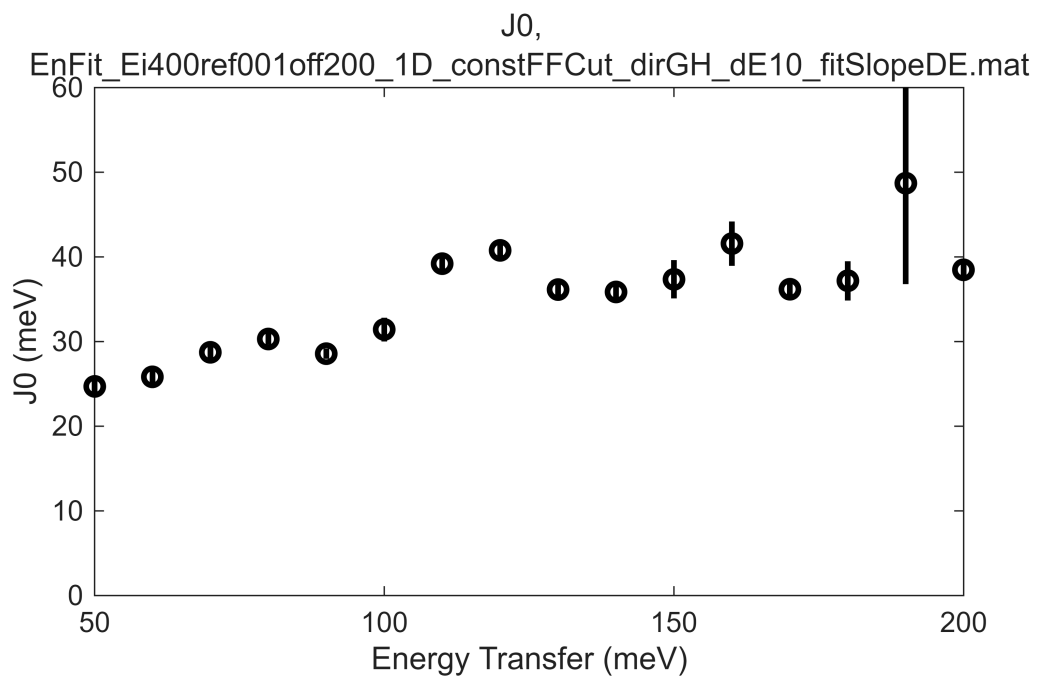
$-0.1 \leq \xi \leq 0.1$ in $[2+\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, \eta, 0]$, $195 \leq E \leq 2$

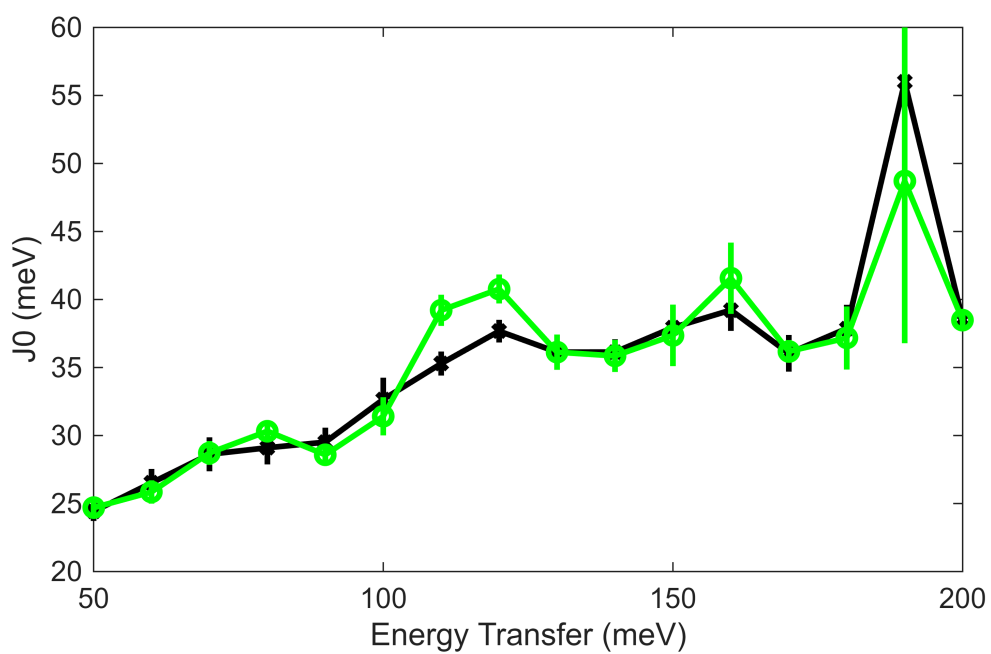
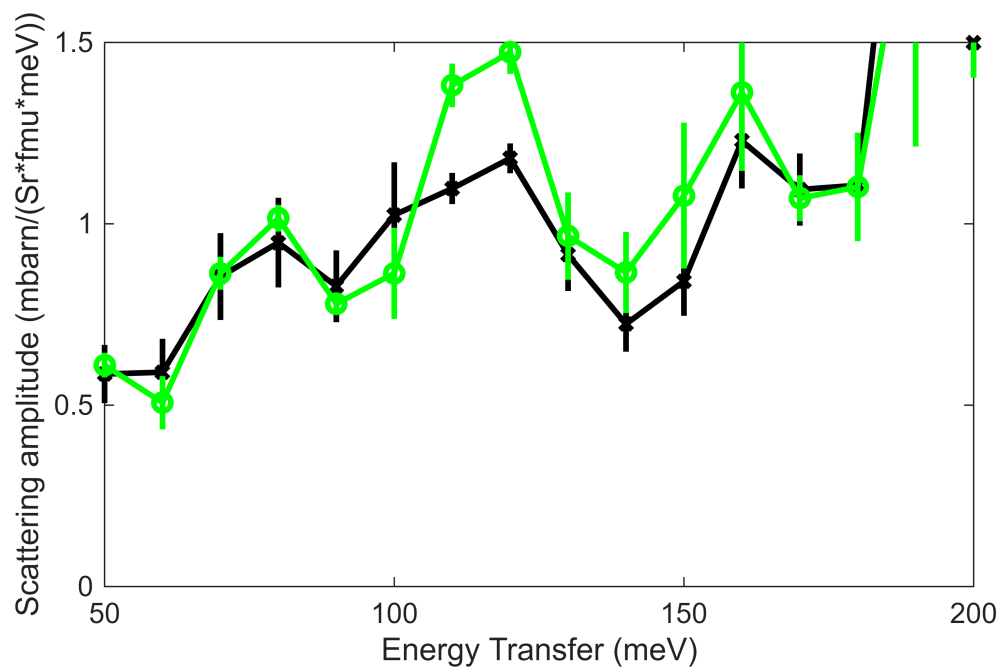
$\zeta = -0.005:0.02:1.015$ in $[2, 0, \zeta]$

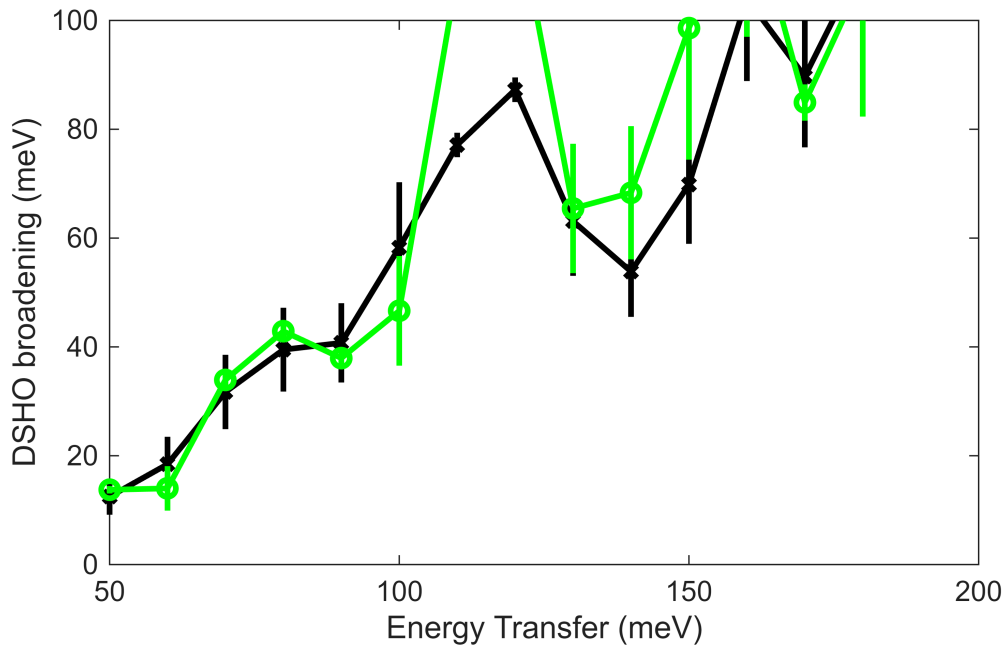












Warning: Error in state of SceneNode.

String scalar or character vector must have valid interpreter syntax:

Amplitude,

e:\SHARE\Fe\Data\analysis\06_fit\with\J0\sym4D\cutsAndFits\EnFit\Ei400ref001off200_2D_constFFCut_dirGH_dE10_fitSlopeDE.mat

Warning: Error in state of SceneNode.

String scalar or character vector must have valid interpreter syntax:

J0,

e:\SHARE\Fe\Data\analysis\06_fit\with\J0\sym4D\cutsAndFits\EnFit\Ei400ref001off200_2D_constFFCut_dirGH_dE10_fitSlopeDE.mat

Warning: Error in state of SceneNode.

String scalar or character vector must have valid interpreter syntax:

Gamma,

e:\SHARE\Fe\Data\analysis\06_fit\with\J0\sym4D\cutsAndFits\EnFit\Ei400ref001off200_2D_constFFCut_dirGH_dE10_fitSlopeDE.mat

```
%cl,'En_cuts800',dir_name,cut_en,dE_step,half_dE);
```

```
beep
```

```
beep
```

```
beep
```

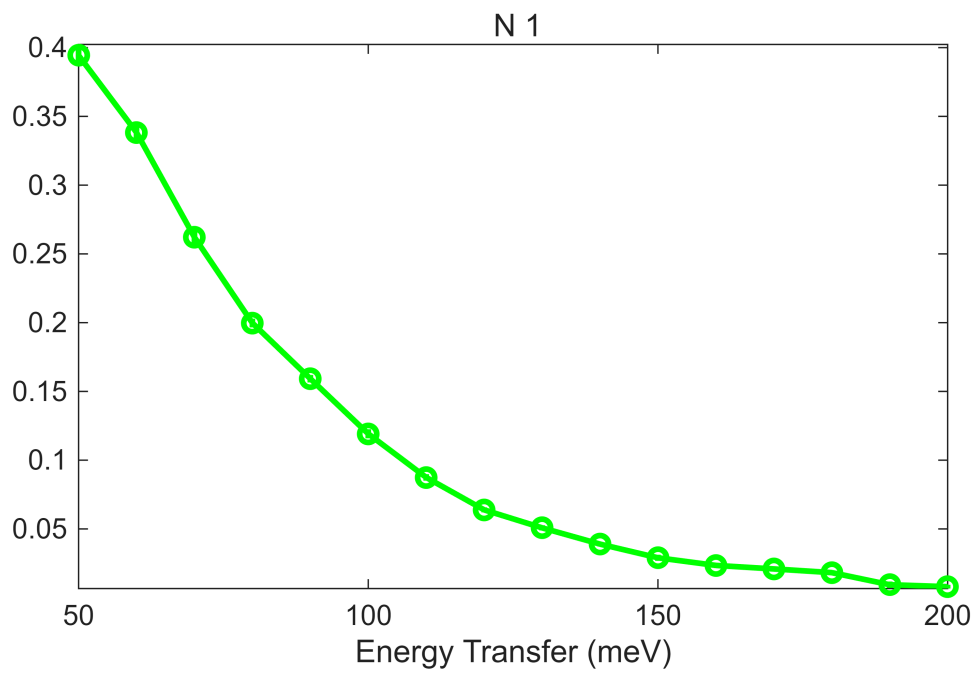
```
fit_res_400 = fit_res_400rec;
```

```
bg_data = extract_bg_par(fit_res_400.all_fit_par,[1,2]);
```

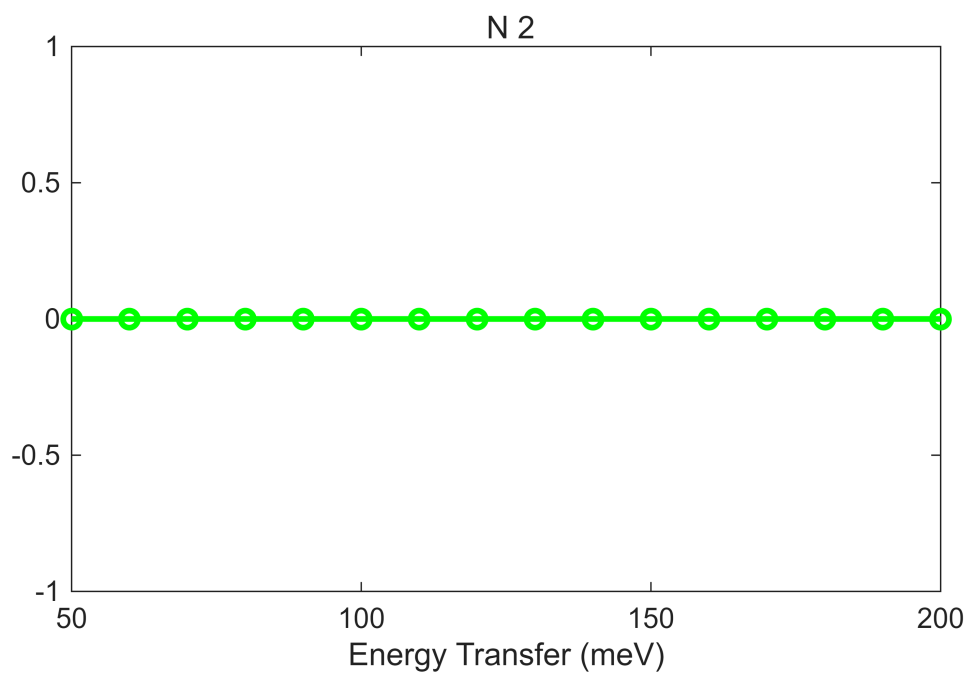
```
% original slope parameters
```

```
pd(bg_data(1));lx(fit_res_400.all_fit_par(1).en,fit_res_400.all_fit_par(end).en);
```

```
keep_figure;
```



```
pd(bg_data(2));lx(fit_res_400.all_fit_par(1).en,fit_res_400.all_fit_par(end).en);
keep_figure
```



```
fit_resEi400 = fit_res_400rec;
save(res_name,"fit_resEi400")
```